

Week 3. Homework 1. SQL Basic

Due date: **September 12, 11:59 pm**

Submission Guidelines

- **Dataset:** You must use the provided *IMDB.sqlite* dataset to answer all questions.
- **Code and Output:** Copy your code and attach screenshots of your outputs under each question.
- **Submission Format:** Each student is required to push one Word or PDF file to a designated GitHub repository you create specifically for this week's assignment.
- **Repository Name:** Use the format *Week 3-HW1-LastName*, where *LastName* is your last name.
- **Code Quality:** Ensure your code is organized for easy understanding.
- **AI Usage:** You should make a serious attempt at each problem prior to using an LLM. Remember, you may be tested on SQL knowledge without assistance in a technical interview. If prompting an LLM, I suggest telling it your starting point, your thought process, and asking it to guide you. I've done this learning SQL and it is helpful and I learned more that way. Disclose your usage at the end of the file; you will not be penalized regardless, but this give me some information.

Questions

1. Directors and Their Departments

List the names of all directors along with their respective departments.

Answer:

SELECT name, department

FROM directors;

	name	department
1	James Cameron	Directing
2	Gore Verbinski	Directing
3	Sam Mendes	Directing
4	Christopher Nolan	Directing
5	Andrew Stanton	Directing
6	Sam Raimi	Directing
7	Byron Howard	Directing
8	Joss Whedon	Directing
9	David Yates	Directing
10	Zack Snyder	Directing
11	Bryan Singer	Directing

Execution finished without errors.
Result: 2349 rows returned in 36ms
At line 1:
SELECT name, department
FROM directors;

2. Popular Movies

Find the titles of all movies with a popularity score greater than 5.

Answer:

```
SELECT *
```

```
FROM movies
```

```
WHERE popularity >5;
```

	id	original_title	budget	popularity	release_date
1077	44716	21 Jump Street	42000000	62	2012-03-12
1078	44717	Notting Hill	42000000	28	1999-05-13
1079	44718	Chicken Run	45000000	31	2000-06-21
1080	44719	Along Came Polly	42000000	33	2004-01-16
1081	44720	Boomerang	40000000	6	1992-06-30
1082	44721	The Heat	43000000	37	2013-06-27
1083	44722	Cleopatra	31115000	13	1963-06-12
1084	44723	Here Comes the Boom	0	26	2012-10-11
1085	44724	High Crimes	42000000	14	2002-04-03
1086	44726	The Mothman Prophecies	32000000	14	2002-01-25

```
Execution finished without errors.  
Result: 3376 rows returned in 79ms  
At line 1:  
SELECT *  
FROM movies  
WHERE popularity >5;
```

3. Directors in the Directing Department

Retrieve the names and IDs of directors who are in the 'Directing' department and have a gender value of '2'.

Answer:

```
SELECT name, id, department, gender
```

```
FROM directors
```

```
WHERE gender = 2;
```

	name	id	department	gender
1	James Cameron	4762	Directing	2
2	Gore Verbinski	4763	Directing	2
3	Sam Mendes	4764	Directing	2
4	Christopher Nolan	4765	Directing	2
5	Andrew Stanton	4766	Directing	2
6	Sam Raimi	4767	Directing	2
7	Byron Howard	4768	Directing	2
8	Joss Whedon	4769	Directing	2
9	David Yates	4770	Directing	2
10	Zack Snyder	4771	Directing	2
11	Bryan Singer	4772	Directing	2

Execution finished without errors.
 Result: 1574 rows returned in 7ms
 At line 1:
 SELECT name, id, department, gender
 FROM directors
 WHERE gender = 2;

4. Directors with 'Sam' in Their Name

Find all directors whose names include the word 'Sam'.

Answer:

SELECT name

FROM directors

WHERE name LIKE '%Sam%';

	name
1	Sam Mendes
2	Sam Raimi
3	Sam Fell
4	Sam Weisman
5	Karyn Kusama
6	Sam Taylor-Johnson
7	Sam Miller
8	Sammo Hung
9	Sam Peckinpah
10	Sam Levinson
11	Sam Firstenberg

Execution finished without errors.
 Result: 11 rows returned in 7ms
 At line 1:
 SELECT name
 FROM directors
 WHERE name LIKE '%Sam%';

5. First 10 Movies by Release Date

Show the first 10 movies in the database, sorted by release date in ascending order.

Answer:

```
SELECT title
```

```
FROM movies
```

```
ORDER BY release_date ASC
```

```
LIMIT 10;
```

	title
1	Intolerance
2	The Big Parade
3	Metropolis
4	Pandora's Box
5	The Broadway Melody
6	Hell's Angels
7	A Farewell to Arms
8	42nd Street
9	She Done Him Wrong
10	It Happened One Night

Execution finished without errors.
Result: 10 rows returned in 10ms
At line 1:
SELECT title
FROM movies
ORDER BY release_date ASC
LIMIT 10;

6. 'Star' Movies with High Ratings

List all movies with the word 'Star' in the title and a vote average greater than 7.

Answer:

```
SELECT title, vote_average
```

```
FROM movies
```

```
WHERE (title LIKE '%Star%')
```

```
AND (vote_average >=7);
```

	title	vote_average
1	Star Trek Into Darkness	7.4
2	Star Trek	7.4
3	Star Wars: Episode III - Revenge of...	7.1
4	Stardust	7.1
5	Star Trek: First Contact	7.0
6	Star Trek II: The Wrath of Khan	7.3
7	The Fault in Our Stars	7.6
8	Star Wars	8.1
9	Star Wars: Clone Wars: Volume 1	8.0
10	20 Feet from Stardom	7.4

```

Execution finished without errors.
Result: 10 rows returned in 10ms
At line 1:
SELECT title, vote_average
FROM movies
WHERE (title LIKE '%Star%')
AND (vote_average >=7);

```

7. Movies with Limited Revenue (1995–2000)

Find the titles of movies released between January 1, 1995, and December 31, 2000, with revenues not exceeding 10 million.

Answer:

```
SELECT *
```

```
FROM movies
```

```
WHERE release_date BETWEEN '1995-01-01' AND '2000-12-31'
```

```
AND revenue <= 10000000;
```

	id	original_title	budget	popularity	release_date
1	43973	Volcano	90000000	19	1997-04-25
2	44060	Déjà Vu	0	0	1998-04-22
3	44064	Mighty Joe Young	90000000	6	1998-12-25
4	44068	Little Nicky	85000000	18	2000-11-10
5	44115	Inspector Gadget	75000000	17	1999-07-23
6	44133	Anna and the King	75000000	13	1999-12-16
7	44253	Primary Colors	65000000	9	1998-03-20
8	44257	Proof of Life	65000000	14	2000-12-08
9	44262	Lucky Numbers	65000000	2	2000-10-27
10	44307	The Family Man	60000000	19	2000-12-12

```

Execution finished without errors.
Result: 328 rows returned in 13ms
At line 1:
SELECT *
FROM movies
WHERE release_date BETWEEN '1995-01-01' AND '2000-12-31'
AND revenue <= 10000000;

```

8. Directors with Specific 'uid' Values

List the names of directors who have a uid greater than 7,000 or less than 100.

Answer:

```
SELECT name, uid
```

```
FROM directors
```

```
WHERE uid > 7000 OR uid < 100;
```

	name	uid
1	Sam Mendes	39
2	Andrew Stanton	7
3	Sam Raimi	7623
4	Byron Howard	76595
5	Joss Whedon	12891
6	David Yates	11343
7	Zack Snyder	15217
8	Bryan Singer	9032
9	Marc Forster	12995
10	Rob Marshall	17633
11	Marc Webb	87742

Execution finished without errors.
Result: 2071 rows returned in 8ms
At line 1:
SELECT name, uid
FROM directors
WHERE uid > 7000 OR uid < 100;

9. Movies with Specific Vote Counts

Retrieve the titles of movies that have:

- A vote counts between 50 and 100, inclusive.
- A vote average of exactly 7.0.

Answer:

```
SELECT title, vote_count, vote_average
```

```
FROM movies
```

```
WHERE (vote_count BETWEEN 50 AND 100)
```

```
AND (vote_average = 7);
```

	title	vote_count	vote_average
1	Angela's Ashes	59	7.0
2	A Better Life	54	7.0
3	Red Riding: In the Year of Our Lord...	54	7.0
4	Quo Vadis	63	7.0
5	Another Year	97	7.0
6	Salvador	71	7.0
7	The Black Stallion	59	7.0
8	Elling	51	7.0
9	Fireproof	100	7.0

```

Execution finished without errors.
Result: 9 rows returned in 14ms
At line 1:
SELECT title, vote_count, vote_average
FROM movies
WHERE (vote_count BETWEEN 50 AND 100)
AND (vote_average = 7);

```

10. Directors Named 'John' with Specific 'uid' Values

Find the names of directors who:

- Have 'John' in their name.
- Have a uid between 5,000 and 10,000 (exclusive).

Answer:

SELECT name, uid

FROM directors

WHERE (name LIKE '%John%')

AND (uid BETWEEN 5000 AND 10000);

	name	uid
1	John Lasseter	7879
2	John Milius	8328
3	John Singleton	6482
4	John Guillermin	6778
5	John McNaughton	9200
6	John Ottman	9039
7	John Huston	6593
8	John Ford	8500

```

Execution finished without errors.
Result: 8 rows returned in 7ms
At line 1:
SELECT name, uid
FROM directors
WHERE (name LIKE '%John%')
AND (uid BETWEEN 5000 AND 10000);

```

11. Movies with Odd 'uid' and Even Vote Count

Retrieve the titles of movies that meet the following criteria:

- The uid is an odd number.
- The vote count is an even number and greater than 500.

Answer:

```
SELECT *
```

```
FROM movies
```

```
WHERE uid %2!=0
```

```
AND (vote_count >500 AND vote_count%2=0);
```

	id	original_title	budget	popularity	release_date
199	44717	Notting Hill	42000000	28	1999-05-13
200	44718	Chicken Run	45000000	31	2000-06-21
201	44735	The Interview	44000000	74	2014-12-25
202	44738	8 Mile	41000000	32	2002-11-08
203	44751	Fifty Shades of Grey	40000000	98	2015-02-11
204	44760	Scary Movie 4	45000000	32	2006-04-13
205	44763	Get Hard	40000000	36	2015-03-26
206	44765	Julie & Julia	40000000	22	2009-08-06

Execution finished without errors.
Result: 405 rows returned in 113ms
At line 1:
SELECT *
FROM movies
WHERE uid %2!=0
AND (vote_count >500 AND vote_count%2=0);

12. Movies with Specific Ratings or Budgets

Find the titles of movies that meet any of the following criteria:

- Have a vote average between 6.0 and 8.0, with a vote count of exactly 200.
- OR have a budget greater than 100,000 but less than 500,000.

Answer:

```
SELECT *
```

```
FROM movies
```

```
WHERE (vote_average BETWEEN 6.0 AND 8.0)
```

```
AND (vote_count = 200)
```

```
AND (100000 > budget < 500000);
```

	id	original_title	budget	popularity	release_date	revenue	title	vote_average	vote_co
1	47035	Control	0	11	2007-09-26	8159508	Control	7.5	200
2	47670	Sleeper	2000000	10	1973-12-17	18344729	Sleeper	7.0	200